

NEW YORK STEWART, NY, USA

WMO: 725038

Lat: 41.500N

Lon: 74.100W

Elev: 150

StdP: 99.54

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14714

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.3	-13.4	-22.5	0.5	-13.9	-20.2	0.6	-11.3	12.8	-2.1	11.8	-5.1	3.3	290	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.4	22.3	30.9	22.0	29.0	20.9	24.3	29.3	23.4	28.1	22.6	27.0	4.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	17.8	26.3	22.3	17.2	25.8	21.2	16.2	25.0	73.8	29.4	70.5	28.1	67.1	27.1	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.3	8.5	-20.3	34.9	3.7	2.2	-23.0	36.5	-25.2	37.8	-27.2	39.1	-30.0	40.7		
(5)			-19.7	25.7	2.9	1.5	-21.8	26.8	-23.5	27.6	-25.1	28.4	-27.2	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-3.5	-1.2	3.2	9.4	14.7	20.3	22.8	22.0	17.5	11.0	5.4	0.1		
	DBStd	10.00	6.29	5.30	5.01	4.64	3.90	3.63	2.93	3.03	3.59	4.06	4.63	4.83		
	HDD10.0	1514	419	313	217	64	4	0	0	0	0	37	151	308		
	HDD18.3	3354	677	546	469	270	127	21	2	6	57	228	388	565		
	CDD10.0	1593	1	1	8	47	150	310	396	372	226	69	13	2		
	CDD18.3	391	0	0	1	3	14	81	139	119	33	2	0	0		
	CDH23.3	3311	0	0	7	57	178	673	1186	940	245	24	1	0		
	CDH26.7	1041	0	0	2	17	51	209	413	297	50	2	0	0		
(14) Wind	WSAvg	3.8	4.2	4.0	4.4	4.4	3.9	3.4	3.3	2.9	3.2	3.6	4.0	4.1		
(15) Precipitation	PrecAvg	1111	77	62	86	93	107	109	100	96	102	90	95	81		
	PrecMax	1496	250	116	192	224	258	291	346	243	251	455	194	190		
	PrecMin	719	4	6	10	30	30	10	21	16	19	1	21	21		
	PrecStd	194	45	26	38	38	54	60	59	50	60	70	43	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.2	15.9	23.8	28.9	31.3	32.7	35.3	34.1	30.9	26.1	20.3	17.5		
		MCWB	13.3	11.0	14.9	18.3	19.3	22.9	23.7	22.7	21.2	16.9	15.8	12.2		
	2%	DB	11.3	11.3	17.8	24.1	27.8	31.1	32.6	31.7	27.8	23.0	17.9	12.5		
		MCWB	9.6	5.7	11.0	14.6	18.4	22.4	22.8	22.0	19.6	16.1	14.2	9.6		
	5%	DB	7.6	8.7	14.5	20.2	24.9	29.0	30.9	29.8	26.2	20.9	16.0	9.7		
		MCWB	5.7	5.4	8.8	13.2	16.7	21.4	22.3	21.3	18.7	15.4	11.9	7.3		
	10%	DB	5.1	6.4	11.2	17.6	22.3	27.4	28.8	27.9	24.0	17.9	12.9	7.3		
		MCWB	2.9	2.9	6.9	10.8	14.9	20.4	21.4	20.4	18.2	13.4	9.4	4.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.0	11.8	15.5	19.3	21.4	25.0	26.4	24.9	23.5	20.0	17.2	12.6		
		MCDB	15.5	14.8	22.8	27.7	27.6	29.7	32.8	29.7	26.6	22.7	18.5	15.6		
	2%	WB	10.0	7.9	12.6	15.9	19.7	23.7	24.6	23.8	21.8	17.6	15.4	9.9		
		MCDB	11.1	9.8	16.5	21.8	25.1	28.7	29.7	28.7	25.2	20.9	18.0	12.2		
	5%	WB	6.4	5.2	9.9	14.3	18.1	22.7	23.5	23.0	20.7	15.9	12.6	7.7		
		MCDB	7.7	8.1	14.2	19.2	23.0	27.2	28.2	27.7	23.3	19.4	14.7	9.3		
	10%	WB	3.0	3.1	7.3	12.2	16.6	21.5	22.6	22.2	19.5	14.3	10.3	4.6		
		MCDB	4.9	6.0	10.7	15.9	21.0	25.7	27.1	26.2	22.9	17.5	12.8	6.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.6	9.9	11.4	11.5	10.9	11.1	10.9	11.2	10.9	9.4	8.3		
		MCDBR	10.7	12.9	14.5	16.2	15.9	12.8	13.2	12.9	13.1	15.1	12.3	11.6		
		MCWBR	9.1	9.6	9.5	9.1	8.0	6.3	5.4	5.3	6.4	8.6	9.0	9.4		
		MCWBR	10.4	11.1	13.9	13.9	13.2	11.3	10.9	10.1	10.4	12.3	10.4	10.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.300	0.318	0.344	0.363	0.405	0.464	0.473	0.453	0.381	0.352	0.318	0.301		
		taud	2.418	2.381	2.394	2.392	2.296	2.166	2.154	2.200	2.377	2.470	2.518	2.500		
		Ebn at Noon	852	889	904	908	872	818	806	811	853	842	828	825		
		Edh at Noon	79	96	106	115	129	148	148	137	107	87	72	68		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.78	2.65	3.63	4.62	5.16	5.61	5.83	5.10	4.22	2.83	1.97	1.45		
		RadStd	0.13	0.26	0.28	0.49	0.52	0.53	0.33	0.33	0.44	0.30	0.18	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+109	+60

Nomenclature: See separate page